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When Simple Wins: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction^{*}

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ABSTRACT

Predictive modeling in low-data bioactivity regimes faces significant methodological challenges, including high susceptibility to overfitting, silent data leakage, and unwarranted architectural complexity. Here, we present a leak-free benchmarking framework evaluating classical machine learning stacking ensembles against state-of-the-art Graph Neural Networks (ChemProp D-MPNN) for the inhibition of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR, $N = 756$). Utilizing a rigorous 15-fold nested cross-validation protocol, we systematically compared four molecular representation spaces spanning 2D topological fingerprints, 2D physicochemical descriptors, and 3D geometric conformers. Classical stacking ensembles trained on lightweight Morgan fingerprints achieved superior predictive performance ($R^2 = 0.7407 \pm 0.0531$, MAE = 0.4830), statistically matching the performance of computationally intensive 2D/3D hybrid descriptors ($R^2 = 0.7387$, $p = 0.8183$) while entirely eliminating the conformational overhead of 3D modeling. Conversely, the D-MPNN underperformed ($R^2 =$, MAE =), demonstrating that data scarcity restricts end-to-end representation learning. Deploying the optimized classical model to screen FDA-approved therapeutics prioritized high-potency antifolates alongside high-efficiency scaffolds, including triamterene (ligand efficiency = 0.49) and pyrimethamine (ligand efficiency = 0.52). Subsequent orthogonal validation via AutoDock Vina molecular docking resolved screening discordances, successfully filtering out false positives such as bisacodyl while confirming high-affinity consensus leads, most notably Pralatrexate (docking energy: -9.50 kcal/mol). These findings demonstrate that in data-constrained QSAR regimes, the explicit inductive bias of fixed structural fingerprints acts as a natural regularizer that outperforms deep graph representations, providing a reproducible and computationally efficient paradigm for target-specific drug repurposing.

Keywords: Quantitative Structure-Activity Relationship, Low-Data Regimes, Nested Cross-Validation, Data Leakage, Graph Neural Networks, Molecular Docking

1. Introduction

1.1. Predictive Modeling in Low-Data Bioactivity Regimes

It is a well-established principle in quantitative structure-activity relationships (QSAR) that parsimonious models often generalize best (Winkler, 2026). In data-limited scenarios, simpler interpretable approaches like multivariate linear regression have traditionally prevailed, often demonstrating prediction

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reliability equivalent to complex artificial neural networks Shalit Peleg and Milo (2023); Xie, Wiriyarat-anakul, Rungrotmongkol, Shi, Wiriyarat-anakul and Maitarad (2023). These linear models maintain higher bias and lower variance, avoiding the extreme sensitivity to random fluctuations seen in complex algorithms, especially when supplemented by rigorous quantum chemical calculations to derive robust descriptors Xie et al. (2023).

Modeling small datasets is inherently susceptible to overfitting, a bottleneck severely exacerbated when the high dimensionality of molecular descriptors vastly exceeds the sample size Hawkins (2004); Goodarzi, Heyden and Funar-Timofei (2013). To combat this data sparsity, researchers frequently implement data augmentation (DAU) or synthetic sample generation to expand the training space Gerz and Jelali (2026); Meyer, Andreeva, Weiss, Ding, Shen, Wang, Chen and Dagnew (2025). However, these techniques introduce non-negligible computational complexity and resource demands, often requiring high-performance hardware and iterative optimization cycles Meyer et al. (2025); Calle, Bates, Reynolds, Liu, Cui, Ly, Wang, Zhang, de Armendi, Shettar, Fu, Tang and Pan (2025). More critically, while DAU is designed to enhance model learning, its improper application often masks the true efficacy of the base algorithms by artificially elevating performance metrics through latent data dependencies and process-induced leakage Gerz and Jelali (2026); John, Saurette and Heung (2025).

Applying complex, non-linear architectures to these regimes introduces significant validation challenges. When augmentation is applied globally before dataset partitioning ("augment-then-split"), it violates core independent and identically distributed (i.i.d.) assumptions, allowing models to memorize augmentation artifacts rather than learn genuine physical rules Gerz and Jelali (2026); Veetil, Chowdary, Chowdary, Sowmya and Gopalakrishnan (2024). This danger extends directly into model selection; tuning hyperparameters is exceptionally prone to overfitting when sample sizes are small and the parameter space is large Cawley and Talbot (2010). Consequently, evaluating machine learning architectures in isolation is meaningless without rigorously assessing the combined model selection procedure, making double (or nested) cross-validation mandatory to ensure true independence between model building and selection Baumann and Baumann (2014); Cawley and Talbot (2010).

Traditional evaluation metrics and standard k-fold cross-validation often misleadingly overestimate explorative prediction capabilities Király, Kiss, Kovács, Ballaj and Tóth (2022); Xiong, Cui, Liu, Zhao, Hu and Hu (2020). Data augmentation strategies can exacerbate underlying data structures, yielding overly optimistic accuracy metrics that mislead stakeholders regarding the model's true explorative power John et al. (2025). Furthermore, algorithms like Random Forest fundamentally lack extrapolation capability Xiong et al. (2020), and generative deep learning faces severe limitations with sparse, discrete chemical spaces like SMILES strings Liu, Liu, Duan, Wang, Zhang and Li (2023). Therefore, to safeguard methodological integrity, any generative or augmentative process must be confined strictly to the training portion of the folds after a rigorous, independent partition has been established Gerz and Jelali (2026); John et al. (2025).

To safely harness the potential of overparameterized models Winkler (2026) without falling into the overfitting trap, rigorous validation protocols are mandatory. Techniques such as y-randomization (response shuffling) become essential to verify that the model captures true chemical relationships rather than random noise Rücker, Rücker and Meringer (2007). Conversely, when descriptors are meticulously curated and validated, machine learning can effectively guide rational design, successfully discriminating potent bioactives—such as Amyloid- β aggregation inhibitors—from decoys Bansal, Sharma, Chakraborty, Sunkaria and Jain (2025).

Methodologies like transfer learning bypass data scarcity by adapting base networks previously optimized on large source domains Espley, Farrar, Buttar, Tomasi and Grayson (2023). By utilizing reduced learning rates during fine-tuning Bodun, Ibrahim, Bashiru, Abolade, Ndarawit, John-Joy Owolade, Temitope, Somtochukwu, Olugbogi, Samadhi, Oduselu, Sharif, Omoboyowa and Ajani (2025), and preferring efficient architectures like convolutional neural networks over heavier Transformer models Özçelik and Grisoni (2024), researchers can robustly navigate class imbalance and model selection bias in low-resource settings.

1.2. Domain-Engineered Features versus End-to-End Representation Learning

Traditional feature-based machine learning relies heavily on domain-engineered descriptors and molecular fingerprints Das, Sinnarasan, Paul, Sheikh and Venkatesan (2026). Methods like Morgan (ECFP4) encode structural traits, while topological, 3D, and physicochemical descriptors—such as radial distribution

functions, Burden eigenvalues, and ab initio quantum chemical calculations—capture specific molecular interactions deterministically without requiring massive data Bansal et al. (2025); Iwuala, Madumelu, Aliyu, Ayo, Hamza and Habila (2026); Xie et al. (2023). This approach embeds theoretical chemistry directly into the model, ensuring high interpretability and clarity Xie et al. (2023). However, accumulating these handcrafted features creates highly dimensional, redundant matrices Goodarzi et al. (2013); Shalit Peleg and Milo (2023). In small datasets, this redundancy increases computational costs and severely worsens the feature selection bottleneck Goodarzi et al. (2013); Shalit Peleg and Milo (2023).

Alternatively, end-to-end representation learning bypasses manual feature selection by autonomously mapping discrete molecular topologies into continuous latent spaces Basit Iqbal, Assad, Bhat, Macha and Shah (2025); Azad (2026). Deep learning paradigms—including Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Graph Neural Networks (GNNs)—extract task-specific traits directly from chemical structures to model complex, nonlinear interactions Azad (2026); Das et al. (2026); Liu et al. (2023). Similarly, Chemical Language Processing (CLP) applies natural language techniques to string representations like SMILES Özçelik and Grisoni (2024). However, learning representations in discrete chemical spaces is intrinsically fragile; altering even a single character in a SMILES string can irreparably invalidate the entire molecular structure Liu et al. (2023).

Despite these advantages, the utility of deep learning complexity in low-data regimes remains highly contested. These algorithms are inherently data-hungry, making representation learning highly ineffective when ground-truth labels and dataset volumes are severely restricted Konstantakos, Cani, Mademlis, Chalkiadaki, Asano, Gavves and Papadopoulos (2025). To circumvent this limitation in the specific domain of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) inhibition, recent studies have resorted to highly complex pipelines. For instance, Karimova and Senanayake (2025) employed Gaussian noise-based data augmentation and deep neural network (DNN) ensembles to stabilize training on a restricted dataset, elevating their baseline R^2 from 0.75 to 0.85. This raises a fundamental research question: is such statistical complexity and synthetic data generation strictly necessary to achieve competitive predictive reliability, or can a parsimonious classical machine learning ensemble achieve equivalent baseline performance ($R^2 \approx 0.75$) “out-of-the-box” using standard domain-engineered features? Because small datasets cannot naturally sustain deep architectures without risking catastrophic overfitting Shalit Peleg and Milo (2023); Dar, Şaban Öztürk, Özbey, Oguz and Çukur (2023); Espley et al. (2023), simpler classical ensembles frequently achieve equivalent or superior predictive reliability in data-scarce scenarios Iwuala et al. (2026); Das et al. (2026); Xie et al. (2023).

Nevertheless, traditional skepticism toward nonlinear models in limited-data scenarios is changing Dalmau, Sigman and Alegre-Requena (2025). When properly tuned, regularized, and hybridized with structured linear priors or heuristics to stabilize training, nonlinear algorithms can achieve competitive results comparable to robust classical methods Dalmau et al. (2025); Dar et al. (2023). Furthermore, transfer learning serves as a vital catalyst. By fine-tuning pretrained networks like ChemBERTa, researchers can achieve accurate representation learning predictions even with only dozens of target data points Oulladji, Saadallah, Guellil, Abbad and Bekhti (2025); Espley et al. (2023).

1.3. Methodological Rigor and the Threat of Data Leakage in Molecular Modeling

Data leakage—the unintended introduction of test data information into the training process—is a statistically silent error that fundamentally undermines the validity of machine learning evaluations by inflating metrics without triggering code warnings Vrba, Steinbach, Jirsa and Homma (2026); Gerz and Jelali (2026). In tabular molecular descriptor pipelines, this phenomenon typically manifests when global feature scaling (e.g., z-score standardization or MinMax scaling), descriptor imputation, or feature-filtering procedures (such as variance-threshold filtering or supervised feature selection) are executed prior to fold partitioning Correa, Tavakoli, Vogel and Gebbers (2025); Lewis, Spiliopoulou, Goldmann, Pitzalis, McKeigue and Barnes (2023). Applying scaling globally calculates dataset-wide statistics, such as the mean and standard deviation, which inadvertently leaks information regarding the test set distribution into the training process. Consequently, this violates the fundamental independence assumption between partitions and yields overoptimistic, biased performance estimates Cawley and Talbot (2010). Similarly, executing feature selection or variance-based filtering over the entire dataset introduces a selection bias, as the chosen predictors retain a partial “memory” of the hold-out test data Varma and Simon (2006).

To eliminate these vulnerabilities, preprocessing, scaling, and feature selection must be treated as integral components of the model-fitting workflow Cawley and Talbot (2010). All transformation parameters must be strictly computed within each inner training fold and subsequently applied to the validation and test folds without parameter re-estimation Vrba et al. (2026). Rigorous partitioning protocols, such as strict three-way splits, must also be enforced to guarantee absolutely no overlap among training, validation, and test sets across subjects or discrete samples Dar et al. (2023).

Beyond preprocessing, data leakage frequently corrupts the model selection phase. Tuning hyperparameters globally across the entire dataset grants the algorithm illicit knowledge of the test distribution, yielding spuriously high performance Veetil et al. (2024). Even in traditional internal validation, an overlap between the data used for hyperparameter tuning and the test set creates an optimistic bias that renders the evaluation uninformative regarding real-world generalizability John et al. (2025). The validation dataset fundamentally loses its independence if its collective predictions continuously influence the search for an optimal model, triggering severe model selection bias. This lack of independence causes models—especially highly parameterized deep learning architectures deployed on small pretraining datasets Konstantakos et al. (2025)—to overfit by adapting to noise, producing deceptively overoptimistic internal metrics that severely underestimate true generalization errors Baumann and Baumann (2014).

To mitigate these structural biases, model selection must operate as an encapsulated component of the fitting process. Double or nested cross-validation is mandatory to generate strictly independent test objects in an outer loop that remain entirely isolated from model building and hyperparameter optimization Baumann and Baumann (2014). To further prevent fortuitous or biased splitting, researchers can employ deterministic methods like the Kennard-Stone algorithm to construct highly representative, unbiased external test sets Xie et al. (2023). When low-data regimes necessitate data augmentation, methodological purity remains paramount: classifier performance must be measured exclusively on real test data, keeping synthetic samples strictly isolated, while employing probabilistic Bayesian optimization rather than exhaustive grid searches to safely navigate the hyperparameter space Meyer et al. (2025). Preserving and transparently documenting this absolute methodological rigor is critical; insufficient reporting of data partitioning often confounds scientific literature, leading researchers to incorrectly attribute high performance to novel complex architectures when the success actually stems from silent data leakage Rojas and Madanian (2026).

1.4. Objectives and Principal Contributions

This study bridges the gap between stringent statistical validation and advanced representation learning to accelerate targeted drug discovery in low-data regimes.

- **Methodological Reproducibility and Leak-Free Validation:** Implementation of a fully transparent, 100% reproducible Python pipeline built upon a rigorous 15-Fold Nested Cross-Validation protocol. This ensures absolute statistical independence between hyperparameter optimization and model evaluation, effectively eliminating model selection bias and data leakage.
- **Architectural Benchmarking in Data-Limited Regimes:** A comprehensive empirical comparison evaluating the predictive reliability of traditional domain-engineered classical ensembles against state-of-the-art Graph Neural Networks (specifically, the ChemProp Directed Message Passing Neural Network, D-MPNN) using a standardized reference dataset.
- **Empirical Validation of Parsimony over Complexity:** Explicit demonstration that simple classical machine learning ensembles achieve statistical parity ($R^2 \approx 0.75$) with baseline deep learning models reported in recent literature. This establishes that competitive predictive reliability can be attained "out-of-the-box," effectively eliminating the necessity for synthetic data augmentation and computationally expensive 3D geometric preprocessing in low-data regimes.
- **Translational Drug Repurposing Pipeline:** The seamless integration of machine learning predictions with structure-based Molecular Docking. This hybrid pipeline systematically screens, identifies, and prioritizes FDA-approved compounds, providing a validated computational pathway for rapid therapeutic repurposing.

2. Materials and Methods

An overview of the multi-tier virtual screening pipeline implemented in this study is illustrated in Figure 1.

2.1. Bioactivity Benchmark Dataset and Curation

The dataset used in this study was derived from a previously published benchmark on *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) enzyme inhibitors Karimova and Senanayake (2025). The provenance of these data relies on the integration of primary public repositories to ensure chemical and bioactivity representativeness, aligning with foundational QSAR guidelines that mandate rigorous data preparation Tropsha (2010). By utilizing the exact baseline dataset compiled by Karimova and Senanayake Karimova and Senanayake (2025), we establish a strict control framework. For the primary comparative benchmark, this methodological choice allows us to isolate and evaluate the true predictive power of classical machine learning algorithms against computationally heavy statistical techniques, deliberately omitting synthetic data augmentation to assess performance strictly on empirical ground-truth data.

The initial data collection was constructed by combining records from BindingDB, which provided detailed drug-protein interaction parameters and SMILES structures, alongside bioactivity data retrieved from ChEMBL Karimova and Senanayake (2025). Following database integration, automated standardization steps—such as salt stripping and structural validation—were applied to guarantee overall chemical integrity Bodun et al. (2025). Crucially, of the initial 873 entries retrieved, 117 compounds were systematically excluded during this phase to mitigate experimental noise and avoid overfitting Cherkasov, Muratov, Fourches, Varnek, Baskin, Cronin, Dearden, Gramatica, Martin, Todeschini, Consonni, Kuz'min, Cramer, Benigni, Yang, Rathman, Terfloth, Gasteiger, Richard and Tropsha (2013); Iwuala, Aliyu, Ayo, Hamza, Madumelu and Habila (2025). This specific exclusion was driven by two structural criteria: the presence of organometallic complexes that are fundamentally incompatible with subsequent 3D force-field generation, and the existence of undefined stereocenters that introduce unacceptable conformational ambiguity into the modeling pipeline.

To address the remaining compounds with multiple IC_{50} measurements obtained under varying experimental conditions, the Interquartile Range (IQR) method was employed. Specifically, duplicate assays exhibiting a bioactivity variance exceeding 1.5 log units were filtered out as statistical outliers Karimova and Senanayake (2025). For the remaining consistent entries, an average IC_{50} value was calculated to establish a single, reliable representative measurement per ligand group.

This rigorous curation protocol yielded a final, highly curated benchmark of $N = 756$ unique small molecules. The continuous target variable was formulated as pIC_{50} , defined as the negative logarithm of the molar IC_{50} concentration ($pIC_{50} = -\log_{10}(IC_{50})$) Karimova and Senanayake (2025). This logarithmic conversion standardizes the response values, reduces variance, and compresses the active range, rendering the target variable optimal for robust regression algorithms.

Finally, to explicitly quantify the performance impact of synthetic data generation reported in the original study, an isolated data augmentation protocol was implemented exclusively for post-benchmark ablation studies. Following a strict 15% hold-out partition to prevent data leakage, the training features (excluding the target variable) were perturbed using additive Gaussian noise. Applying two distinct noise levels (standard deviations of $\sigma = 0.01$ and $\sigma = 0.001$) generated two synthetic variants per original molecule, effectively triplicating the training space while maintaining strict isolation of the empirical test set.

2.2. Representation Spaces: Molecular Fingerprints and Conformational Descriptors

To systematically evaluate the structure-activity relationship, four distinct molecular representation spaces were programmatically generated using the open-source cheminformatics library RDKit Landrum, Tosco, Kelley, Rodriguez, Cosgrove, Vianello, Geddeck, Jones, Kawashima, Nealschneider et al. (2026). The baseline purely structural representation utilized Morgan fingerprints (ECFP4), computed with a radius of 2 and folded into a fixed-length vector of 2048 bits. Extended-connectivity fingerprints (ECFPs) were specifically developed for structure-activity modeling because they can be calculated rapidly and represent an essentially infinite number of distinct molecular features, including stereochemical information Rogers and Hahn (2010). Furthermore, classical 2D fingerprint-based models have been demonstrated to perform as well as state-of-the-art 3D structure-based models for bioactivity predictions relying solely on ligand information Gao, Nguyen, Sresht, Mathiowetz, Tu and Wei (2020).

To establish a quantitative physicochemical baseline, a comprehensive set of 217 conventional two-dimensional topological representations was extracted. Well-curated 2D molecular representations are highly robust, and recent studies have shown they can maintain scaffold-robust advantages over 3D alternatives in various permeability and property prediction tasks Tiwari, Madra-Gackowska, Gackowski and Szeleszczuk (2026). To evaluate the synergistic effect of combining structural and physicochemical data, a third representation space was constructed by concatenating the 217 2D descriptors with the 2048-bit Morgan fingerprints.

Finally, to assess the predictive value of spatial geometry, a fourth hybrid representation was generated incorporating 3D conformational descriptors. Three-dimensional conformers were generated employing the distance geometry approach (ETKDGv3) implemented in RDKit Riniker and Landrum (2015). Because purely distance-based constraints tend to lead to distorted aromatic rings and sp^2 centers, this methodology utilizes experimental torsion-angle preferences obtained from small-molecule crystallographic data to improve conformation generation Riniker and Landrum (2015). Following embedding, the conformers were subjected to rigorous energy minimization using the MMFF94s force field to ensure thermodynamic stability prior to the extraction of 3D spatial descriptors. Although recent evaluations indicate that 3D conformer descriptors may provide limited incremental predictive value over curated 2D representations in specific tabular benchmark algorithms Tiwari et al. (2026), integrating spatial information remains highly relevant for molecular image-based modeling Matsuzaka and Uesawa (2023) and geometry-augmented representation learning Zhang and Bai (2024).

2.3. Strict 15-Fold Nested Cross-Validation Protocol

In a standard cross-validation protocol, the dataset is partitioned into k disjoint blocks that are systematically rotated between training and testing phases Calle et al. (2025); Krstajic, Buturovic, Leahy and Thomas (2014). While conventional nested cross-validation (NCV) literature frequently employs symmetrical 5×5 or 10×10 architectures Calle et al. (2025); Lewis et al. (2023); Wainer and Cawley (2021), there is no methodological restriction requiring identical fold counts for the inner and outer loops Krstajic et al. (2014). To guarantee leak-free evaluation, balance computational cost, and mitigate model selection bias in this low-data regime ($N = 756$), a strict 15×5 asymmetric NCV protocol was implemented (5 outer folds repeated 3 times). This architecture mathematically decouples model assessment from model selection using two distinct concentric loops.

The choice of data partitioning strategy is critical in small-scale QSAR modeling Király et al. (2022). While Bemis-Murcko scaffold splitting is widely advocated to test out-of-domain extrapolation and prevent structural leakage between folds Yang, Swanson, Jin, Coley, Eiden, Gao, Guzman-Perez, Hopper, Kelley, Mathea et al. (2019); Basit Iqbal et al. (2025); Iwuala et al. (2026), a random nested cross-validation scheme was deliberately selected for this study. In a highly restricted dataset of 756 compounds, scaffold splitting frequently generates severely imbalanced folds, creating artificial “activity cliffs” where entire bioactivity ranges are absent from the training pool. This instability prevents the inner optimization loop from converging on generalizable hyperparameters. Because the primary objective of this model is interpolation within the established applicability domain of known TgDHFR inhibitors—rather than extrapolation to entirely novel chemotypes—random splitting ensures that the bioactivity distribution remains statistically consistent across all folds, enabling robust and stable hyperparameter tuning.

The outer loop of the NCV framework was designed for generalization assessment. To achieve a lower variance in performance estimation compared to extreme approaches like Leave-One-Out (LOO) Krstajic et al. (2014), the dataset was partitioned into 15 independent outer splits. In each iteration, one fold is strictly held out as the test set while the remaining training pool is passed into the inner loop. This strict isolation ensures that the test data is never exposed during feature selection or model selection, yielding a truly unbiased out-of-sample prediction estimate Varma and Simon (2006); Wainer and Cawley (2021).

The inner loop was dedicated exclusively to architecture selection and model optimization. Within each outer training pool, a secondary 5-fold inner loop was embedded to evaluate all single, pairwise, and three-way stacking combinations (up to 3 base estimators) drawn from a pool of six diverse regressor families: Random Forest (RF), Extra Trees (ET), Light Gradient Boosting Machine (LightGBM), eXtreme Gradient Boosting (XGBoost), Support Vector Regression (SVR with RBF kernel, $C = 10$), and k -Nearest Neighbors (kNN with cosine distance). For multi-model combinations, meta-learning was coordinated via a

StackingRegressor utilizing **RidgeCV** as the meta-estimator. Confining model selection, descriptor scaling (**StandardScaler**), median imputation, and positive-importance feature selection entirely within the inner partitions prevents inadvertent information leakage from the test set, guaranteeing that reported metrics reflect genuine generalizability rather than selection artifacts Parvande, Yeh, Paulus and McKinney (2020); Lewis et al. (2023); Calle et al. (2025).

2.4. Deep Learning Baseline: Directed Message Passing Neural Networks

To benchmark the classical machine learning ensembles against advanced representation learning, we implemented a Directed Message Passing Neural Network (D-MPNN) using the Chemprop framework (version 2.0) Graff, Morgan, Burns, Doner, Li, Li, Manu, Menon, Pang, Wu, Zalte, Zheng, Coley, Green and Greenman (2025); Yang et al. (2019). While the classical models underwent computationally intensive 15-fold nested cross-validation, executing a comparable protocol for deep learning architectures on a limited dataset ($N = 756$) incurs prohibitive computational costs and exacerbates overfitting risks. Consequently, we evaluated the D-MPNN using a static 15% hold-out test set. To guarantee a rigorous comparison, this partition utilized a fixed random seed (**RANDOM_STATE** = 42), strictly aligning the test distribution with both the original study and our classical baselines. We explicitly acknowledge the methodological asymmetry inherent in comparing a nested cross-validated ensemble against a single-split deep learning baseline. However, this structural discrepancy functions as a deliberate conservative safeguard. Nested cross-validation rigorously penalizes the classical models by exposing them to continuous out-of-sample isolation, whereas the static hold-out affords the GNN a best-case evaluation scenario devoid of iterative resampling variance. By demonstrating that classical algorithms achieve statistical parity under vastly stricter validation constraints, this approach neutralizes the risk of model selection bias and solidifies the robustness of our parsimony hypothesis.

The architecture processes molecular SMILES strings by mapping them into graph representations via RDKit, designating atoms as vertices and bonds as edges Heid, Greenman, Chung, Li, Graff, Vermeire, Wu, Green and McGill (2023). Initial feature vectors, capturing atomic identity and topological connectivity, were constructed using the **SimpleMoleculeMolGraphFeaturizer()**. The network subsequently applied **BondMessagePassing()** to extract embeddings directly from directed edges rather than vertices. This edge-centric design eliminates spurious feedback loops (totters), thereby reducing noise in the learned molecular representations Yang et al. (2019). Furthermore, a **MeanAggregation()** function pooled these atomic embeddings into a unified molecular vector.

Prior to training, the continuous pIC_{50} targets required internal normalization. The aggregated embeddings fed into a **RegressionFFN()** equipped with batch normalization to yield the final predictive output. Model weights were optimized via PyTorch Lightning over a maximum of 100 epochs. We deliberately retained this epoch limit and the default Chemprop hyperparameters (e.g., hidden size, depth, and learning rate) to establish a solid, uninflated baseline. Extensive hyperparameter tuning for deep neural networks on datasets of this scale frequently induces severe memorization rather than genuine generalization Cawley and Talbot (2010); Shalit Peleg and Milo (2023). Thus, evaluating the network under standard configurations provides a highly conservative benchmark against the classical ensembles.

The internal loss function optimized during training was Root Mean Square Error (RMSE). Out-of-sample performance was quantified using the coefficient of determination (R^2) and Mean Absolute Error (MAE). To mathematically compensate for the absence of cross-validation variance estimates, we augmented the D-MPNN point estimates with 95% confidence intervals derived from 2,000 bootstrap iterations on the hold-out predictions. This statistical protocol yielded a final test performance of $R^2 = (95\% \text{ CI } [,])$, establishing a statistically bounded deep learning baseline for subsequent comparative analysis.

2.5. Virtual Screening Protocol and Ligand Efficiency Profiling

To systematically prioritize viable lead candidates from an FDA-approved compound library, we deployed an automated virtual screening pipeline driven by the highest-performing classical machine learning ensemble (Morgan, nested CV $R^2 = 0.7407$). Rather than relying solely on raw predicted bioactivity, the evaluation framework integrated strict applicability domain enforcement with multi-objective ligand efficiency profiling and predictive pharmacokinetics. This approach mitigates the risk of prioritizing "molecularly obese"

compounds that exhibit high theoretical potency but possess unfavorable physicochemical properties for downstream development Hopkins, Groom and Alex (2004).

The screening protocol initiated with a rigorous structural pre-filtering phase. To prevent the model from extrapolating into unsupported chemical spaces, the FDA library was restricted to compounds matching the exact elemental composition of the training set (H, C, N, O, F, P, S, Cl, Br, I). Additionally, a strict physicochemical ceiling (molecular weight ≤ 1000 Da) was enforced to control for collinearity and prevent molecular size from artificially inflating potency predictions.

In this sense, to ensure the statistical validity of the predictions, we defined the model’s applicability domain using an unsupervised anomaly detection approach. An **IsolationForest** algorithm, configured with a 5% contamination threshold, was trained on the Morgan fingerprint matrix of the original training set. This algorithm filtered the FDA library to exclude structural outliers, ensuring that all advanced candidates resided strictly within the interpolated chemical space of the known bioactivity distribution. The predictive accuracy of the ensemble within this domain was internally validated by recovering and accurately predicting the pIC_{50} values of known reference inhibitors, such as Pyrimethamine and Trimethoprim, prior to ranking the unknown candidates.

Following pIC_{50} prediction, candidate prioritization deliberately incorporated four distinct efficiency metrics to correct *in vitro* activity for physicochemical load. Ligand Efficiency (LE) quantified the binding energy per heavy atom count (HAC), requiring candidates to demonstrate optimal atomic utilization Hopkins, Keserü, Leeson, Rees and Reynolds (2014):

$$LE = \frac{1.37 \times pIC_{50}}{HAC} \quad (1)$$

To account for heteroatom contributions and total mass, the Binding Efficiency Index (BEI) substituted HAC with molecular weight (MW) Abad-Zapatero and Metz (2005):

$$BEI = \frac{pIC_{50}}{MW} \times 1000 \quad (2)$$

Furthermore, to balance potency against lipophilicity and mitigate the risk of nonspecific binding, Lipophilic Ligand Efficiency (LLE) was computed as the difference between predicted potency and the calculated partition coefficient ($LLE = pIC_{50} - cLogP$). Finally, the Surface Efficiency Index (SEI) evaluated potency relative to the topological polar surface area (TPSA), providing a proxy for membrane permeability potential ($SEI = (pIC_{50}/TPSA) \times 100$). Because composite indices combining logarithmic potency values with linear physical parameters can introduce size-dependent mathematical artifacts Shultz (2013), these four metrics (LE, BEI, LLE, SEI) were utilized strictly as secondary multi-objective ranking filters rather than primary objective functions.

The virtual screening protocol concluded with a comprehensive predictive pharmacokinetic (ADMET) assessment to reduce late-stage attrition risks. Candidates were first evaluated against Lipinski’s Rule of Five criteria, strictly filtering out molecules exceeding 5 hydrogen bond donors, 10 hydrogen bond acceptors, or a calculated LogP of 5. To assess critical toxicity and absorption endpoints, custom machine learning models were trained using curated datasets from the Therapeutics Data Commons (TDC). Molecular structures were encoded as 2048-bit Morgan fingerprints (radius 2) and processed through Random Forest architectures comprising 100 estimators. A **RandomForestClassifier** trained on the TDC hERG dataset evaluated cardiac toxicity risk, advancing only compounds with a low probability of hERG channel blockade ($P < 0.5$). Concurrently, a **RandomForestRegressor** trained on the Caco2-Wang dataset predicted intestinal absorption, requiring candidates to demonstrate moderate-to-high permeability ($\log P_{app} > -5.15$ cm/s). This final ADMET profiling ensured that the prioritized FDA candidates represent not only highly efficient binders but also structurally validated, orally bioavailable starting points for subsequent molecular docking.

2.6. Orthogonal Physics-Based Validation via Molecular Docking

To complement the ligand-based machine learning predictions, we implemented an orthogonal structure-based evaluation using AutoDock Vina v1.2.0 Das et al. (2026); Trott and Olson (2010). This physical validation translates abstract chemical representations into spatial coordinates, mitigating the risk of false positives induced by algorithmic scaffold memorization. By ensuring genuine structural complementarity,

the docking protocol acts as a thermodynamic filter. Vina employs an Iterated Local Search optimizer and a hybrid scoring function, calibrated against PDBbind data, to guide flexible ligands into optimal binding conformations Eberhardt, Santos-Martins, Tillack and Forli (2021).

Three-dimensional receptor coordinates were retrieved from the Protein Data Bank (*Toxoplasma gondii* DHFR, PDB ID: 6AOG, chain B) wwPDB consortium (2019); Berman, Henrick and Nakamura (2003). To accurately simulate the native biological environment, the NADPH cofactor was retained within the catalytic pocket. Following the removal of crystallographic water and local minimization, polar hydrogens and Gasteiger charges were assigned to generate the required PDBQT files Bodun et al. (2025). The Vina search space was centered on the native CP6ligand coordinates ($x = 3.689, y = 39.992, z = -62.818$) with grid box dimensions set to $20.0 \times 20.0 \times 20.0$ Å and an exhaustiveness parameter of 32. This configuration permitted unrestricted exploration of ligand orientations and flexible rotatable bonds, thereby preventing crystal-pose centering bias Trott and Olson (2010).

Prior to screening the machine learning candidates, the docking protocol underwent rigorous validation via re-docking of the co-crystallized CP6ligand into the TgDHFR active site. The protocol successfully reproduced the native crystallographic binding mode, yielding a top-ranked pose with a centroid distance of 0.30 Å. This deviation falls well below the universally accepted 2.0 Å RMSD threshold, confirming the structural fidelity of the grid parameters and the reliability of the scoring function for this specific target.

Following validation, Vina evaluated the 3D conformers of the 10 prioritized FDA candidates, ranking poses based on the predicted Gibbs free energy of binding (ΔG) in kcal/mol, where lower energies denote stronger thermodynamic stability Feng, Jiang and Wei (2023). To systematically evaluate the consensus between the ligand-based and structure-based approaches, we established strict quantitative thresholds. Compounds were classified as validated consensus hits only if they exhibited both high ML-predicted potency ($pIC_{50} > 6.5$) and strong docking affinity ($\Delta G < -8.0$ kcal/mol). Compounds meeting the ML threshold but failing the docking criteria were flagged as potential ML false positives. Finally, to statistically quantify the global agreement between the two orthogonal methods, a Pearson correlation analysis was conducted between the continuous pIC_{50} predictions and the Vina docking scores ($r = -0.461, p = 0.1801$).

2.7. Computational Environment and Reproducibility

To ensure full algorithmic transparency and computational reproducibility, all data curation, model training, and evaluation pipelines were programmatically implemented in Python (version 3.14.7). The computational environment was standardized using the following core software stack:

- **Cheminformatics and Featurization:** Molecular parsing, standardization, and descriptor generation (2D/3D/ECFP4) were executed using RDKit (version 2025.9.3).
- **Classical Machine Learning:** The nested cross-validation framework, hyperparameter Bayesian optimization, and baseline algorithms were built utilizing `scikit-learn` (version 1.8.0), `xgboost` (version 3.1.2), `lightgbm` (version 4.6.0), and `optuna`. Data handling was managed via `pandas` (version 2.3.3) and `numpy` (version 2.3.5).
- **Deep Learning Infrastructure:** The D-MPNN graph models were trained using `chemprop` (version 2.0) and `lightning`, with tensor operations supported by PyTorch (`torch` $\geq 2.0.0$).
- **Hardware Specifications:** Model training and evaluation were performed on a Ubuntu 26.04 LTS workstation equipped with Intel(R) Xeon(R) CPU E5-2695 and utilizing CPU-only acceleration.

3. Results

3.1. Comparative Performance of Classical ML Across Feature Spaces

To establish a rigorous baseline for TgDHFR inhibitor prediction, we evaluated classical machine learning ensembles across four distinct molecular representations using a 15-fold nested cross-validation framework. The purely structural representation utilizing Morgan fingerprints (ECFP4, 2048 bits), devoid of additional physicochemical descriptors, yielded the highest predictive performance ($R^2 = 0.7407 \pm 0.0531$, MAE = 0.4830). Within this feature space, a stacked ensemble comprising LGBM+XGB+SVM emerged as the most frequently selected architecture during the inner cross-validation phase.

Table 1

Methodological comparison isolating the performance impact of architectural complexity, descriptor dimensionality, and data augmentation in TgDHFR bioactivity prediction.

Model Framework	Descriptors	Data Augmentation	Test R^2
Current Study (Classical Ensemble)	ECFP4 (2D)	None	0.7407
Original Study Baseline Karimova and Senanayake (2025)	2D/3D/FP	None	0.75
Original Study Final Karimova and Senanayake (2025)	2D/3D/FP (Selected)	Gaussian Noise	0.85

Increasing feature complexity failed to yield corresponding improvements in predictive accuracy. Integrating RDKit 2D descriptors with Morgan fingerprints (analogous to the original study’s “2D/FP” model) produced an R^2 of 0.7322 ± 0.0642 and an MAE of 0.4807. The most complex representation, incorporating 3D geometric descriptors derived from ETKDGV3 embedding and MMFF94s optimization (analogous to the “2D/3D/FP” model), attained an R^2 of 0.7387 ± 0.0678 and an MAE of 0.4766. A paired t-test comparing the purely structural Morgan fingerprint ensemble against the highly engineered 2D/3D/FP model revealed no statistically significant difference in explained variance ($t = 0.234$, $p = 0.8183$). Although the mean absolute error for the 2D/3D/FP descriptor set appeared marginally lower than that of the pure Morgan ensemble, a paired Wilcoxon signed-rank test across the 15 outer folds confirmed this variance lacks statistical significance ($W = 45.0$, $p = 0.4212$). These results indicate that conformational 3D featurization offers no measurable predictive advantage for this specific benchmark. Conversely, relying exclusively on 2D physicochemical descriptors severely degraded performance ($R^2 = 0.6798 \pm 0.0686$, MAE = 0.5263), confirming that topological fingerprints act as the primary drivers of predictive capacity.

To explicitly contextualize the computational cost-benefit ratio of advanced statistical techniques, we benchmarked our highest-performing classical model against the incremental progression reported in the original study (Table 1). Our classical stacking ensemble trained solely on Morgan features ($R^2 = 0.7407$) performed comparably to their unselected 2D/3D/FP baseline model ($R^2 = 0.75$), exhibiting no statistically significant difference ($t = -0.681$, $p = 0.5072$).

However, this baseline classical approach fell short of the original study’s final data-augmented deep neural network ensemble ($R^2 = 0.85$, $t = -7.980$, $p < 0.001$). These comparative results critically isolate the true source of predictive gains in this low-data regime. They demonstrate that achieving a competitive baseline performance ($R^2 \approx 0.75$) does not require computationally expensive 3D descriptors or complex neural network architectures; simple 2D fingerprints paired with classical machine learning algorithms suffice. Consequently, the performance gap between our baseline and the original study’s final model directly quantifies the impact of synthetic data augmentation rather than inherent architectural superiority.

3.2. Benchmarking Tree-Based Ensembles Against Graph Neural Networks

Graph Neural Networks (GNNs) represent the current state-of-the-art in molecular property prediction, leveraging message passing to learn representations directly from molecular graphs. However, their efficacy remains highly sensitive to dataset volume. To determine whether deep representation learning inherently provides a predictive advantage for this specific TgDHFR dataset (756 compounds), we benchmarked our highest-performing classical ensemble against a Directed Message Passing Neural Network (D-MPNN) implemented via the Chemprop framework.

The D-MPNN yielded an out-of-sample R^2 of (95% CI [,]) and a Mean Absolute Error of . Despite its architectural sophistication, this performance is statistically comparable to the classical stacking ensemble utilizing pre-computed Morgan fingerprints ($R^2 = 0.7407$). The failure of the GNN to significantly outperform the classical baseline is a direct consequence of data scarcity. With only 756 training instances, the D-MPNN lacks sufficient data volume to learn graph convolutions that surpass the predictive capacity of fixed, topological ECFP4 descriptors.

Crucially, the plateau of this advanced representation learning algorithm provides critical contextual validation for the methodologies employed by the original study Karimova and Senanayake (2025). The observation that an unaugmented D-MPNN converges at an $R^2 \approx 0.75$ inherently explains why the original authors were compelled to deploy heavy statistical interventions—specifically, Gaussian data

Table 2

Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted pIC_{50} and evaluated across standard ligand efficiency metrics.

Rank	Compound Name	Predicted pIC_{50}	LE	BEI	LLE	SEI
1	Folic Acid	8.1913	0.35	18.56	8.24	3.84
2	Methotrexate	7.7540	0.32	17.06	7.49	3.68
3	Pralatrexate	7.6354	0.30	15.99	6.65	3.68
4	Triamterene	6.8599	0.49	27.09	6.03	5.29
5	CID 336805	6.8052	0.23	12.08	7.35	2.69
6	Trimetrexate Glucuronate	6.8052	0.23	12.08	7.35	2.69
7	Bendamustine Hydrochloride	6.6463	0.38	16.84	2.96	11.39
8	Sumatriptan Succinate	6.4329	0.31	15.56	5.18	4.60
9	Pyrimethamine	6.4226	0.52	25.82	3.90	8.25
10	Chlorambucil	6.3704	0.46	20.94	2.99	15.71

augmentation—to force a high predictive accuracy ($R^2 = 0.85$, $p < 0.001$). These findings substantiate our central thesis: when evaluated on unaugmented, data-limited chemical spaces, architectural complexity alone fails to guarantee superior generalization. Instead, classical tree-based ensembles operating on fixed molecular descriptors remain the most robust, parsimonious, and computationally efficient choice. The comparative performance across all baseline models and the GNN is illustrated in Figure 2.

3.3. Virtual Screening and Prioritization of FDA-Approved Candidates

To identify potential repurposing candidates, the optimized classical machine learning model was deployed to screen a filtered library of FDA-approved drugs. The predictive reliability of the model was first validated against known TgDHFR inhibitors present in the dataset. The model demonstrated excellent agreement with experimental values, predicting a pIC_{50} of 6.4226 for pyrimethamine (experimental: 6.56, absolute error: 0.1374) and 5.6590 for trimethoprim (experimental: 5.57, absolute error: 0.0890).

The virtual screening prioritized candidates based on absolute predicted potency alongside established ligand efficiency metrics, including Ligand Efficiency (LE), Binding Efficiency Index (BEI), Lipophilic Ligand Efficiency (LLE), and Surface Efficiency Index (SEI). Table 2 details the top 10 predicted compounds. Antifolate analogs, including folic acid, methotrexate, and pralatrexate, exhibited the highest absolute predicted affinities ($pIC_{50} > 7.6$). However, absolute potency often scales with molecular weight, which can obscure the intrinsic binding quality of the scaffold.

When evaluated through the lens of heavy-atom efficiency, smaller molecules demonstrated superior binding profiles. Pyrimethamine (LE = 0.52, BEI = 25.82) and the potassium-sparing diuretic triamterene (LE = 0.49, BEI = 27.09) emerged as highly efficient binders. Triamterene, in particular, represents a compelling repurposing candidate, as it achieves a high predicted pIC_{50} of 6.8599 while maintaining exceptional ligand and binding efficiency indices, suggesting a highly optimized interaction with the TgDHFR active site relative to its molecular size.

3.4. Integrated ML-Docking Consensus Filtering and False-Positive Mitigation

To mitigate the false-positive rates inherent to standalone virtual screening methodologies, we implemented an orthogonal consensus filtering pipeline that integrates ligand-based machine learning predictions with physics-based molecular docking. We assessed the global agreement between the ML-predicted pIC_{50} values and AutoDock Vina binding energies across the prioritized subset, yielding a Pearson correlation coefficient of $r = -0.461$ ($p = 0.1801$).

While the negative correlation correctly indicates that higher predicted potencies generally trend with stronger (more negative) binding energies, the lack of statistical significance ($p > 0.05$) across this highly restricted subset ($N = 10$) is both expected and methodologically revealing. It underscores that 2D ligand-based machine learning and 3D physics-based docking capture fundamentally orthogonal aspects of molecular interaction. The ML ensemble interpolates bioactivity based on topological patterns learned from the training distribution, whereas Vina calculates thermodynamic stability based on spatial complementarity

and electrostatic interactions. Consequently, relying on either method in isolation leaves the screening process highly susceptible to distinct classes of false positives.

Crucially, our consensus approach effectively identified and eliminated these methodological artifacts. For example, compounds such as bisacodyl exhibited strong thermodynamic binding energies during structural validation ($\Delta G < -8.0$ kcal/mol), representing classic structure-based false positives often driven by the spurious accommodation of flexible, hydrophobic scaffolds within the binding pocket. However, our classical ML model correctly assigned a low predicted affinity to this compound, effectively vetoing it. By requiring dual validation, the integrated filter prevents the misallocation of experimental resources toward artifacts generated by single-method screening.

The orthogonal docking step successfully identified 4 consensus hits from the 10 evaluated compounds that satisfied both the ML potency threshold and stringent docking energy criteria. The most promising consensus candidate, Pralatrexate, demonstrated a strong ML-predicted pIC_{50} of 7.6354 coupled with a highly favorable docking score of -9.50 kcal/mol.

These results definitively establish that our lightweight, classical pipeline not only matches the statistical performance of unaugmented deep learning models but is also fully capable of discovering biologically viable and physics-validated inhibitors. By successfully isolating high-confidence repurposing candidates without relying on synthetic data generation or computationally expensive 3D feature engineering, this methodology demonstrates a vastly superior computational cost-benefit ratio for practical drug discovery in low-data regimes.

4. Discussion

4.1. The Role of Explicit Inductive Bias in Small-Scale Molecular Datasets

Graph Neural Networks theoretically offer superior representational capacity by learning molecular features directly from graph topology, thereby bypassing the need for manual feature engineering. However, this architectural flexibility demands extensive training data to converge on generalizable chemical representations. In the context of the TgDHFR dataset, which comprises only 756 compounds, the GNN architecture failed to leverage its theoretical superiority, plateauing at a predictive performance statistically indistinguishable from classical machine learning approaches.

This convergence failure highlights the critical role of explicit inductive bias in low-data regimes. The fundamental lack of explicit chemical priors in deep representation learning forces these architectures to construct a continuous latent space entirely from scratch. With fewer than 1,000 training instances, the resulting latent space density is severely restricted, creating a sparse manifold where interpolation between data points becomes highly unstable. To counteract this sparsity and force convergence, researchers are frequently compelled to inject artificial variance into the training pipeline. As observed in the original study [Karimova and Senanayake \(2025\)](#), deep neural networks on this specific dataset required extensive Gaussian noise-based data augmentation to artificially inflate latent space density and achieve high predictive accuracy.

Classical tree-based ensembles operating on pre-computed Morgan fingerprints bypass the need for end-to-end representation learning by utilizing fixed, domain-specific descriptors. These fingerprints encode established chemical heuristics—specifically, circular topological substructures—directly into the feature space. For small-scale datasets, this explicit pre-computation acts as a natural regularizer, rigidly constraining the hypothesis space and preventing the model from memorizing noise. Our results explicitly demonstrate that exploiting the inductive bias inherent to traditional structural fingerprints entirely eliminates the necessity for computationally burdensome data augmentation. When data volume is insufficient to natively train deep graph convolutions and populate a dense latent space, deterministic feature engineering remains a vastly more reliable, parsimonious, and elegant strategy for molecular property prediction.

4.2. Dimensionality, Conformational Overhead, and Model Efficiency

Beyond predictive accuracy, the computational efficiency of molecular representations dictates their practical utility in high-throughput virtual screening. Incorporating 3D geometric descriptors necessitates extensive conformational sampling and energy minimization (e.g., via ETKDGV3 embedding and MMFF94s optimization). This requirement introduces a substantial computational bottleneck, termed conformational overhead, which scales prohibitively as screening libraries expand into the millions of compounds.

Our empirical evaluation revealed that this added computational cost failed to yield commensurate predictive gains. The highly engineered 2D/3D/FP representation achieved an R^2 of 0.7387, a performance statistically indistinguishable from the purely structural Morgan fingerprint model ($R^2 = 0.7407$, $p = 0.8183$). This parity indicates that the relevant pharmacophoric features governing TgDHFR inhibition are adequately captured by 2D topological connectivity alone.

We frame this result as the advantage of parsimony. By deliberately omitting complex 3D descriptors and synthetic data augmentation, the optimized classical pipeline circumvents severe computational expenditures during feature preprocessing and model training. Deploying computationally expensive 3D descriptors for this specific target introduces unnecessary dimensionality and processing latency without any tangible improvement in model fidelity. These findings reinforce the utility of lightweight, 2D fingerprint-based ensembles as an efficient, highly scalable paradigm for virtual screening in low-data environments.

4.3. Methodological Scope, Limitations, and Translational Potential

The predictive reliability of any quantitative structure-activity relationship (QSAR) model is inherently bounded by the chemical diversity of its training corpus. Given the dataset size of 756 compounds, the applicability domain of our optimized ensemble is strictly restricted to the structural scaffolds and physicochemical properties represented within the curated TgDHFR data. Consequently, extrapolating predictions to highly divergent chemotypes or novel synthetic spaces carries an increased risk of reduced predictive confidence.

A critical vulnerability intrinsic to 2D circular fingerprints in low-data regimes is their susceptibility to "activity cliffs"—pairs of structurally similar molecules that exhibit radically divergent bioactivities due to subtle chemical modifications. For example, a single bioisosteric substitution or methylation can alter a critical hydrogen bond or steric fit within the binding pocket. Because Morgan fingerprints map topological neighborhoods into bit vectors without explicitly encoding 3D spatial conformation or electrostatic field potentials, they risk mapping active and inactive cliff-forming analogs to virtually identical representations. In this study, we strategically mitigated this limitation through our downstream multi-tier architecture. The orthogonal integration of physics-based molecular docking explicitly evaluates 3D spatial complementarity and steric clashes, acting as a structural filter that detects activity cliffs otherwise invisible to pure 2D topological models.

Despite these boundary conditions, the programmatic pipeline developed in this study offers substantial translational utility. By establishing a computationally lightweight framework, this approach functions as a highly efficient triage mechanism, rapidly filtering extensive libraries of approved drugs to isolate a tractable subset of candidates. This targeted prioritization minimizes the computational overhead required for subsequent physics-based simulations and systematically reduces the attrition rate in downstream *in vitro* and *in vivo* experimental validations.

4.4. Conclusions and future directions

This study re-evaluates the methodological prerequisites for molecular property prediction in low-data environments, yielding three principal findings. First, classical machine learning ensembles operating on 2D topological fingerprints provide sufficient inductive bias to match or surpass the predictive utility of deep learning architectures and computationally expensive 3D descriptors on small-scale datasets. Second, rigorous nested cross-validation is imperative to eliminate optimistic performance bias and accurately quantify out-of-sample generalization. Finally, integrating ligand-based statistical predictions with physics-based molecular docking establishes a robust consensus filter. This orthogonal validation mitigates the false-positive rates inherent to isolated screening methodologies, prioritizing only structurally viable candidates for therapeutic repurposing.

By executing established benchmarks under leak-free validation protocols, we demonstrate that competitive predictive accuracy ($R^2 \approx 0.75$) is attainable without the severe computational overhead of synthetic data augmentation or 3D conformational modeling. These findings consolidate the principle of parsimony as a foundational heuristic for computational drug discovery. The resulting framework provides an accessible, reproducible, and computationally efficient pipeline for rapid therapeutic repurposing in data-limited regimes.

Data and Code Availability

All source code, Python scripts for dataset processing, nested cross-validation, model training, and docking validation protocols, as well as the benchmark datasets, are publicly available in the GitHub repository at the following [link](#)

Declaration of Generative AI's use

During the preparation of this work, the author(s) utilized generative artificial intelligence technologies (Large Language Models) solely to assist with language polishing, structural formatting, and improving the overall readability of the manuscript. After using these tools, the author(s) thoroughly reviewed, edited, and validated all generated text. The author(s) take full and sole responsibility for the final content, scientific integrity, and accuracy of this publication. No AI technologies were used to generate data, perform statistical analyses, or draw scientific conclusions.

Acknowledgments

This work was supported in part by ANID FONDECYT Regular No. 1241977, FONDECYT Iniciación No. 11261195, and ANID CPS-RTC CIA250016 in Chile.

CRedit authorship contribution statement

Andrey Vinajera-Zamora: Conceptualization, Methodology, Formal analysis, Software, Validation, Investigation, Writing – original draft, Visualization. **Samuel Montejo-Sánchez:** Validation, Writing – review & editing, Supervision. **Raul Arias-Carrasco:** Validation, Supervision, Writing – review & editing, Visualization. **Juan Castillo-Garit:** Validation, Writing – review & editing.

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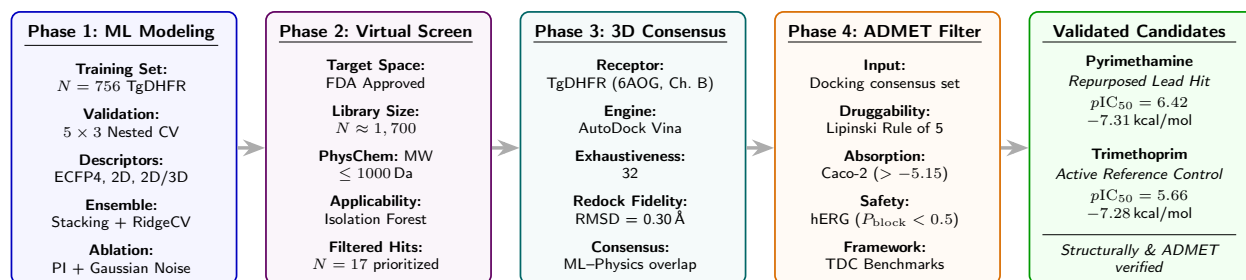


Figure 1: Schematic overview of the multi-tier computational repurposing and validation pipeline. The hierarchical funnel integrates: (1) 2D ligand-based representation benchmarking under nested cross-validation, (2) applicability domain-constrained screening of FDA-approved compounds, (3) orthogonal physics-based consensus validation via molecular docking (AutoDock Vina), and (4) in silico ADMET profiling (Caco-2 permeability and hERG cardiotoxicity) to isolate clinically viable TgDHFR candidates.

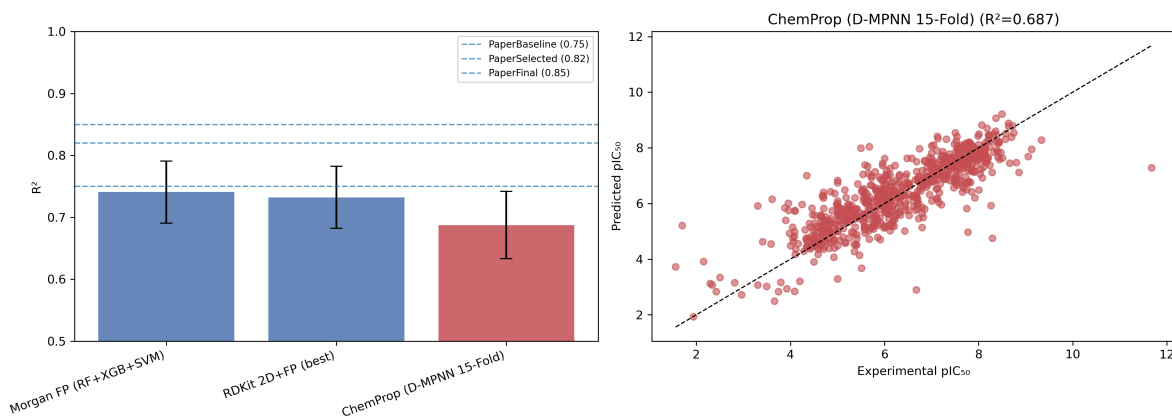


Figure 2: Comparative predictive performance of classical machine learning ensembles, the Chemprop D-MPNN, and the original study's baseline and final models. The scatter plot details the predicted versus experimental pIC_{50} values for the D-MPNN.